

Memoryless property

Given a list of

observations sampled from an arbitrary continuous random variable

, and two numbers

and

check for the memoryless property approximately, i.e. compute

and

from the empirical distribution (the data provided).

Hint: The absolute difference between the previous probabilities should be smaller than a threshold for samples drawn from the exponential distribution and should be big for samples drawn from a different distribution.

Tasks:

1. [4 test cases] Compute
2. .
3. [4 test cases] Compute the conditional probability
4. .
5. [4 test cases] Return True iff
6. .

In order to solve tasks 1., 2. and 3., in file `solution.py` implement functions `prob_larger_than_t`, `prob_larger_than_s_plus_t` and `has_memoryless_property`, respectively.

Input / Output examples

Expected at the command line are

- the number of samples
-
- a non-negative number
-
- a non-negative number
-
- a small positive threshold
-
- a string to select a particular distribution: `exp (`
- `)`, `normal (`
- `)` or `uniform (`
- `)`.
- For the exponential distribution `<dist_param_0>` stands for
- and `<dist_param_1>` should be omitted.
- For the normal distribution `<dist_param_0>` and `<dist_param_1>` stands for the mean
- and standard deviation

- , respectively.
- For the uniform distribution `<dist_param_0>` and `<dist_param_1>` stands for the lower and upper bounds,
- and
- , of the sampling interval, respectively.

Example Input

```
Enter <n> <s> <t> <eps> <dist> <dist_param_0> <dist_param_1>
1000 0.2 0.5 0.02 exp 1
```

Example Output

```
P(X> 0.5) = 0.627
P(X> 0.2 + 0.5 | X> 0.2) = 0.620
||P(X> 0.2 + 0.5 | X> 0.2) - P(X> 0.5)|| < eps = True ...
```

Example Input. (Fails due to low number of samples)

```
Enter <n> <s> <t> <eps> <dist> <dist_param_0> <dist_param_1>:
200 0.2 0.5 0.02 exp 1
```

Example Output

```
P(X> 0.5) = 0.675
P(X> 0.2 + 0.5 | X> 0.2) = 0.593...
||P(X> 0.2 + 0.5 | X> 0.2) - P(X> 0.5)|| < eps = False ...
```

Example Input

```
Enter <n> <s> <t> <eps> <dist> <dist_param_0> <dist_param_1>
1000 2 3 0.01 uniform 0 10
```

Example Output

```
P(X> 3.0 ) = 0.721
P(X> 2.0 + 3.0 | X> 2.0 ) = 0.64
||P(X> 2.0 + 3.0 | X> 2.0 ) - P(X> 3.0 )|| < eps = False ...
```

Example Input

```
Enter <n> <s> <t> <eps> <dist> <dist_param_0> <dist_param_1>
1000 2 3 0.01 normal 5 2
```

Example Output

```
P(X> 3.0 ) = 0.831
P(X> 2.0 + 3.0 | X> 2.0 ) = 0.520...
||P(X> 2.0 + 3.0 | X> 2.0 ) - P(X> 3.0 )|| < eps = False ...
```

Always submit partial solutions

Do not hardcode, we will exchange the test values.